

Agglomerative Hierarchical Clustering (Bottom-up HC)

1. Initialization: Each data point is treated as a separate cluster. For n data points, there are n clusters at the beginning.
2. Distance Calculation: Calculate the pairwise distances between all clusters using a chosen distance metric & linkage criteria.
3. Merge clusters: Identify the two closest clusters based on the distance matrix and merge them into a single cluster.
4. Update Distance Matrix: After merging, update the distance matrix to reflect the distances b/w the new cluster and the remaining clusters.
5. Repeat: Repeat steps 3 & 4 until only one cluster is remaining or until a stopping criteria is reached.

eg. Consider a small dataset of four data points:

Data Point	Feature 1 (x1)	Feature 2 (x2)
A	1	2
B	1	4
C	2	2
D	4	3

Perform Agglomerative Hierarchical clustering using Euclidean distance and single linkage until single cluster is left.

Sol.

Distance	A	B	C	D
A	0	$\sqrt{4}$	$\sqrt{1}$	$\sqrt{10}$
B	$\sqrt{4}$	0	$\sqrt{5}$	$\sqrt{10}$
C	$\sqrt{1}$	$\sqrt{5}$	0	$\sqrt{5}$
D	$\sqrt{10}$	$\sqrt{10}$	$\sqrt{5}$	0

Smallest Distance : $D(A, C) = 1$

Merge A & C : $\{A, C\}$

$\{A, C\}$, $\{B\}$, $\{D\}$

Distance	$\{A, C\}$	B	D
$\{A, C\}$	0	$\sqrt{4}$	$\sqrt{5}$
B	$\sqrt{4}$	0	$\sqrt{10}$
D	$\sqrt{5}$	$\sqrt{10}$	0

Smallest Distance: $D(\{A, C\}, B) = \sqrt{4}$

Merge $\{A, C\}$ and B : $\{A, B, C\}$

$\{A, B, C\}$, $\{D\}$

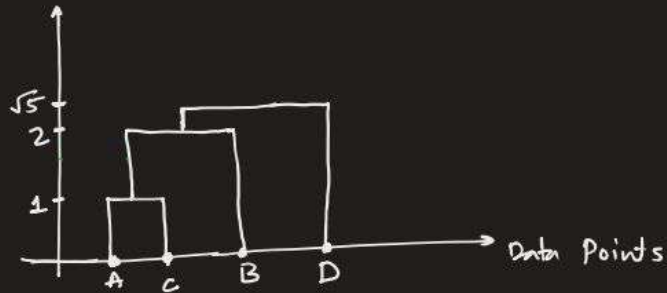
Distance	{ABC}	D
{ABC}	0	$\sqrt{5}$
D	$\sqrt{5}$	0

Smallest Distance: $D(\{ABC\}, D) = \sqrt{5}$

Merge {ABC}, D

: {ABCD}

Distances



Dendrogram

The dendrogram is a tree-like structure that is used to store each step of hierarchical clustering. Leaves of Dendrogram represent data points or observations in the dataset.

Branches connect the leaves and represent the distance or dissimilarity between clusters.

Height at which two clusters are joined reflects the distance between them. A shorter branch means a smaller distance.

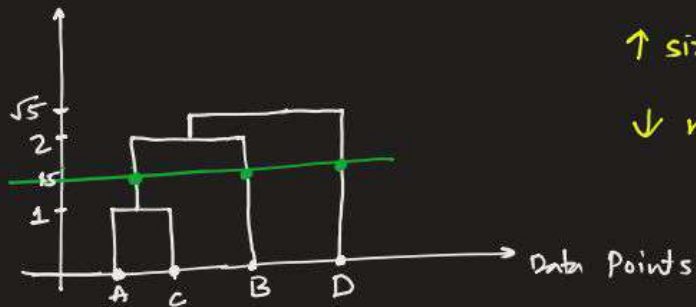
Cutting the Dendrogram: A horizontal line can be drawn across the dendrogram at a certain height to get a certain number of clusters

Distances

↑ height of cutoff

↑ size of clusters

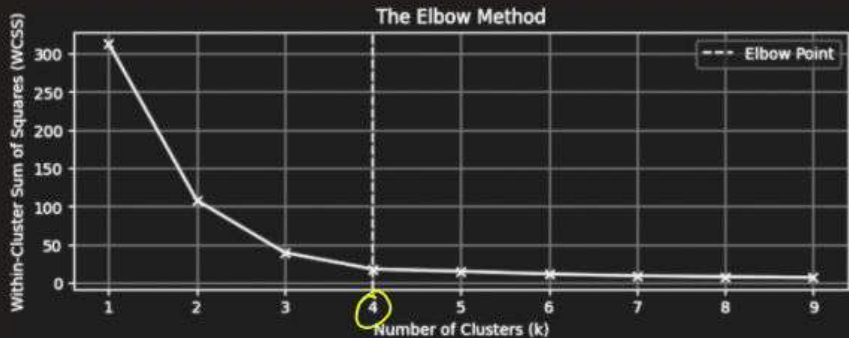
↓ number of clusters



> Elbow method: (optimal value of k)

For different values of k (number of clusters), plot the within cluster sum of squares, then identify the elbow point in the curve to find the optimal value of k .

$K \uparrow$ $WCSS \downarrow$



Before the elbow:

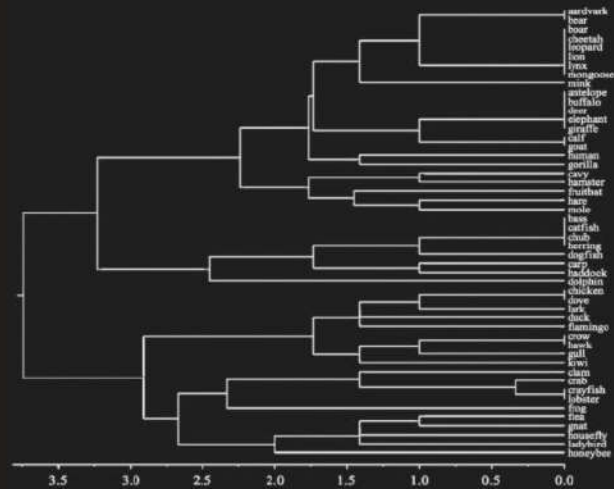
Increase in k significantly reduces $WCSS$.

After the elbow:

Adding more clusters results in a minimal reduction in $WCSS$.

single linkage

↳ distance b/w
clusters & data points is similar



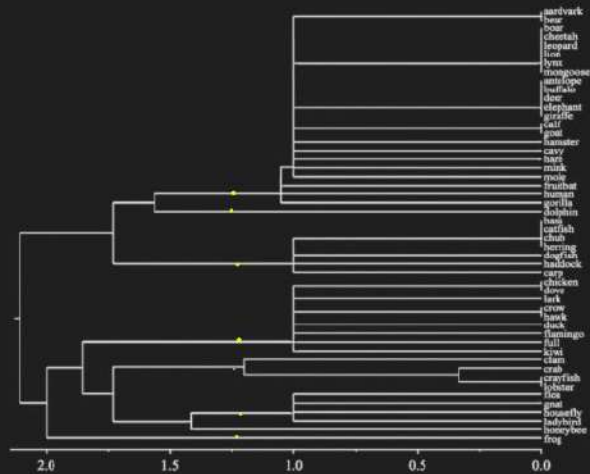
①

Complete Linkage

produces compact clusters at
lower levels of the tree

Complete linkage

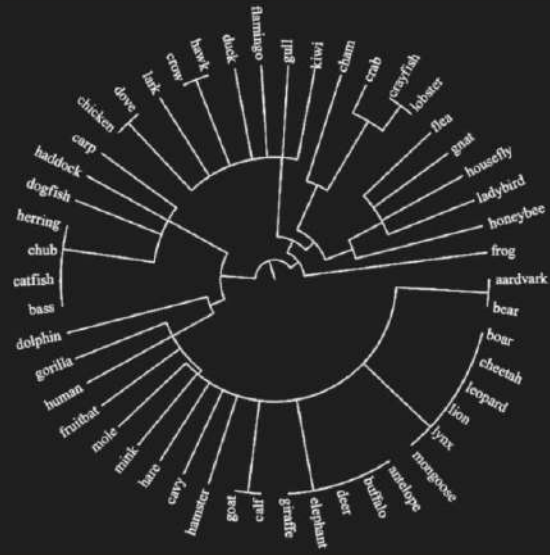
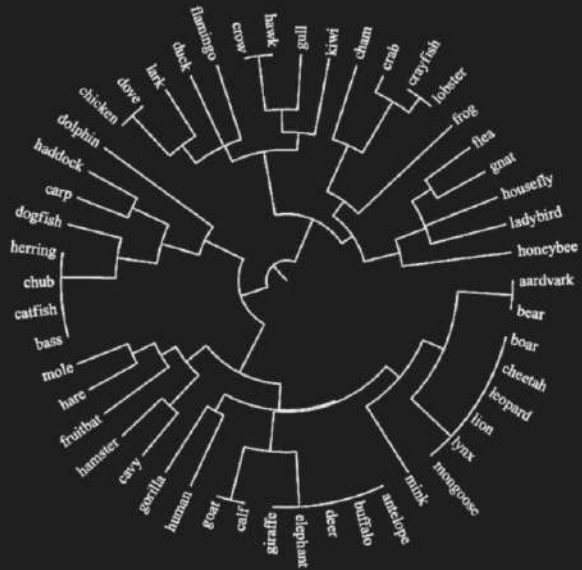
↳ distance b/w clusters ↑ data points ↓



②

Single linkage

produces clusters with large
diameters at low levels of
tree



eg. # Consider the following dataset in 2-D space:

Point	Feature 1	Feature 2
A	1	2
B	2	3
C	3	3
D	5	8
E	8	9

Using Complete linkage and Manhattan distance as distance metric, perform AHC and draw the dendrogram.

Sol.

Distance	A	B	C	D	E
A	0	2	3	10	14
B	2	0	①	8	12
C	3	1	0	7	11
D	10	8	7	0	4
E	14	12	11	4	0

min. distance : $D(B, C) = 1$

$\{BC\}, A, D, E$

Distance	A	$\{BC\}$	D	E
A	0	③	10	14
$\{BC\}$	3	0	8	12
D	10	8	0	4
E	14	12	4	0

Distance	$\{ABC\}$	D	E
$\{ABC\}$	0	10	14
D	10	0	(4)
E	14	4	0

Distance	$\{ABC\}$	$\{DE\}$
$\{ABC\}$	0	14
$\{DE\}$	14	0

Distance
14 ↑

min dist: $D(\{BC\}, A) = 3$

merge $\{BC\}$ and $\{A\}$

$\{ABC\}$, D, E

min dist: $D(D, E) = 4$

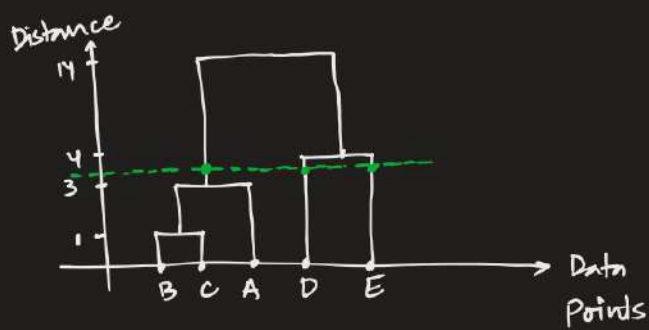
merge D & E

$\{ABC\}$, $\{DE\}$

min dist: $D(\{ABC\}, \{DE\}) = 14$

merge $\{ABC\}$, $\{DE\}$

↳ $\{ABCDE\}$



↳ $\{ABCDE\}$

If we cutoff the dendrogram at 3.5 distance units, how many clusters will be formed?

Ans: (3) $\{ABC\}$, $\{D\}$, $\{E\}$

- Note: We use linkage criteria (min/max/avg/centroid/ward) to calculate the distance matrix only (calculate the distances b/w data points or clusters). After calculating distances, we ALWAYS select min distance clusters (closest clusters) to merge.